Paleomagnetic and petrologic study of the age, origin, and significance of early and late Paleozoic events in the Long Mountain Granite, Wichita Mountains, Oklahoma

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ABSTRACT

The Cambrian Long Mountain Granite, exposed in the western Wichita Mountains, Oklahoma, is red, granophyric, alkali feldspar granite. The red granite abruptly transitions into a green granite in the subsurface; both red and green granites have similar geochemical signatures. Anisotropy of magnetic susceptibility analysis shows that the green granite retains a primary magnetic fabric that is consistent with the sill-like emplacement of the Wichita Granite Group. Demagnetization of green granite specimens yields a characteristic remanent magnetization (ChRM) residing in magnetite with east declinations and moderate down inclinations. The paleopole (9.0°N, 314.6°E) is interpreted as a primary Cambrian thermal remanent magnetization or an early magnetization related to deuteric/hydrothermal alteration. The paleopole position is consistent with some other poles for Cambrian igneous rocks in southern Oklahoma but is to the southwest of the Cambrian part of the apparent polar wander path. The red granite contains abundant secondary hematite that occurs in fractures, as grain boundary coatings, and along cleavage and exsolution planes in alkali feldspars. The iron in the hematite appears to be sourced from the oxidation of primary magnetite and ilmenite and the breakdown of silicate ferromagnesian minerals. The red granite has about two orders of magnitude lower magnetic susceptibility and natural remanent intensity than the green granite. The magnetic fabric is interpreted as an alteration fabric. The ChRM of red granite has southeast declinations and shallow inclinations and is interpreted as a chemical remanent magnetization (CRM) residing in hematite. The paleopole (44.9°S, 304.9°E) falls near the 290–300 Ma segment of the North American apparent polar wander path, which is consistent with inferred timing of exposure of the Long Mountain Granite. The CRM is interpreted to have been acquired during alteration by low-temperature weathering fluids near the surface during the late Paleozoic. The results from the red granite are not consistent with alteration caused by widespread paleoclimatic conditions in the Late Permian, and they are interpreted as related to local tectonic and/or weathering events.

INTRODUCTION

Many granitoid rocks have a red coloration attributed to hematite (e.g., Taylor, 1915; Boone, 1969; Heinrich and Moore, 1970). The origin of the hematite, which commonly occurs as inclusions in feldspars, has been the subject of research for decades. Some workers have proposed a primary magmatic origin for hematite, though this has usually been dismissed on the grounds that other minerals do not contain hematite (e.g., Ernst, 1960). Other workers have suggested solid-state exsolution from iron-bearing feldspar minerals (e.g., Rosenqvist, 1951; Lindqvist, 1966), precipitation during deuteric feldspar recrystallization reactions (e.g., Nakano et al., 2005; Putnis et al., 2007), and hydrothermal activity (e.g., Wenner and Taylor, 1976) as possible causes for the oxide inclusions.

Weathering has also been proposed as a mechanism for red coloration and formation of authigenic hematite found in alteration zones below paleoweathering surfaces in crystalline basement rocks in Europe (e.g., Ricordel et al., 2007; Franke et al., 2010; Fàbrega et al., 2012). These unconformity surfaces contain Permian–Triassic secondary magnetizations, and the red coloration is attributed to weathering fluids that deeply permeated basement rocks and triggered authigenic hematite formation (e.g., Franke et al., 2010; Parcerisa et al., 2010; Ricordel et al., 2007; Parnell et al., 2000). Determining the timing of alteration associated with paleoweathering surfaces can provide a valuable tool for evaluating the geodynamic evolution of ancient continents as well as increasing our understanding of paleoclimate conditions attending oxidation at the paleosurface. Dating paleoweathering surfaces and correlating them with deposits in nearby basins can provide information on continental uplift and basin subsidence (e.g., Franke et al., 2010). In terms of changes in paleoclimate, the Permian-Triassic boundary has been interpreted to represent a shift into Triassic greenhouse conditions and increased weathering (Retallack, 1999; Martin and Macdougall, 1995; Sheldon, 2006). This increased weathering may be recorded in altered granites; testing if this hypothesized event is widespread could improve understanding of the paleoclimatic conditions at the end of the Permian.

Long Mountain, which exposes part of a granitoid intrusion in the western Wichita Mountains, Oklahoma (Fig. 1), provides an ideal place in which to investigate the timing and origin of reddening in granites. Surface exposures of the granite are pink to red, and quarrying operations on the west side of the mountain have exposed gray-green granite at depth. The Wichita Granite Group, of which the Long Mountain Granite is a part, is early Cambrian in age (Ham et al., 1964; Hogan and Gilbert, 1998; Wright et al., 1996). The contrasting exposures provide the opportunity to investigate the primary petrologic and magnetic properties of the rock and changes associated with alteration. Petrologic analysis provides information as to the nature

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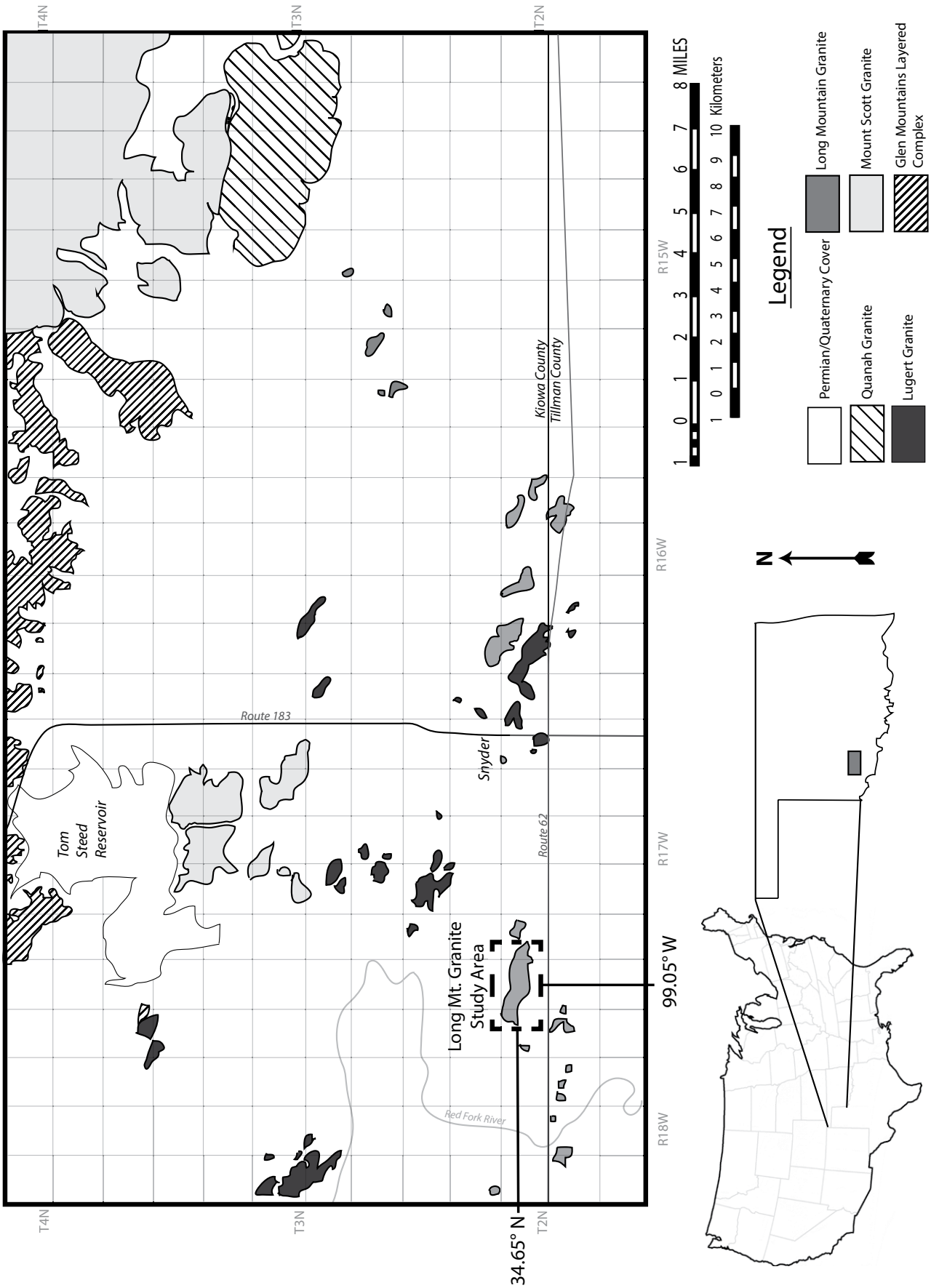


Figure 1. Map of the igneous units in the western portion of the Wichita Mountains, southwestern Oklahoma. The study area is outlined by the dashed rectangle west of Snyder.

of the alteration, and paleomagnetic data can be used to determine the timing of alteration.

The study area also provides an opportunity to obtain a primary Cambrian paleomagnetic pole from the relatively unaltered green Long Mountain Granite. Latest Neoproterozoic to Cambrian paleomagnetism in North America is problematic, with paleopoles of roughly the same age having a wide range of pole positions (e.g., Van der Voo, 1993; McCausland et al., 2007; Abrajevitch and Van der Voo, 2010; Torsvik et al., 2012). This, and the recognition that large numbers of early-obtained paleomagnetic results from Cambrian rocks were completely remagnetized, led to the now-discredited hypothesis of a Cambrian loop in the apparent polar wander path (APWP; e.g., Watts et al., 1980; Lynnes and Van der Voo, 1984) and has led to more modern, yet still controversial, hypotheses such as rapid continental motion due to inertial-interchange true polar wander (Kirschvink et al., 1997) and the presence of an equatorial geomagnetic dipole (Abrajevitch and Van der Voo, 2010; Bono and Tarduno, 2015).

GEOLOGIC BACKGROUND

The Long Mountain Granite is part of the Cambrian igneous province emplaced in the Southern Oklahoma aulacogen. The oldest rocks known to be associated with the aulacogen are those of the Glen Mountains layered complex, a large mafic layered intrusion that forms the backbone of the Wichita Mountains area of the Southern Oklahoma aulacogen (Powell and Phelps, 1977; Hogan and Gilbert, 1998; Hogan et al., 2000; Hanson et al., 2013). Subsequently, large volumes of A-type rhyolites belonging to the Carlton Rhyolite Group were extruded in this area, and series of sheet-like A-type granite sills, including the Long Mountain Granite, were emplaced at the base of the volcanic pile (Hogan and Gilbert, 1995). Following granite intrusion, a series of rhyolitic and diabasic dikes penetrated the section. Once igneous activity ceased, the rift zone began subsiding in the late Cambrian and accumulated sediments until the late Mississippian to early Pennsylvanian, when inversion of the rift occurred during the Ouachita orogeny (Gilbert 1982). During the Pennsylvanian and Permian, igneous rocks were exposed, only to be covered again by Upper Permian sediments. These are now being eroded away, and at present, what is believed to be a fossil Permian surface is being subjected to weathering (Gilbert, 1982).

All extensive granitoids in the Wichita Mountains area are part of the Wichita Granite Group. The granites occur as laterally extensive sheets up to 500 m thick. Surface exposures of the granites are universally red. U-Pb zircon dates from granites and from a rhyolite xenolith in granite range from 535 ± 3 Ma to 530 ± 1 Ma (Wright et al., 1996). U-Pb zircon dates from two rhyolites in the Arbuckle Mountains are 539 ± 5 Ma and 536 ± 5 Ma (Thomas et al., 2012). All Wichita granites have elevated halogen concentrations, and the magmas were water-undersaturated. The felsic magmas are considered to have been intruded or erupted at very high temperatures, in the range of 900–1000 °C (Gilbert, 1982; Hogan and Gilbert, 1997; Hogan et al., 2000). The granitic magmas intruded nonexplosively into the subvolcanic pile at depths of as little as 1 km (Hogan and Gilbert, 1997; Hogan et al., 2000). The elevated halogen content (especially fluorine) would have also greatly lowered the viscosity of the magmas, allowing them to travel large distances (Hogan and Gilbert, 1997; Hogan et al., 2000).

Previous studies of the Long Mountain Granite describe it as a medium-grained, granophyric granite predominantly composed of exsolved feldspars and quartz, with accessory magnetite, titanite (sphene), hornblende, and zircon, and secondary epidote and calcite (e.g., Hessa, 1964). The granite changes to a gray-green color with depth (Hessa, 1964). Taylor (1915) attributed the red coloration to very fine-grained disseminated hematite, which he hypothesized was produced by the alteration of primary minerals via groundwater.

There have been no previous paleomagnetic studies of the Long Mountain Granite, although some studies on the basement rocks in the Wichita Mountains have been published. Several workers reported late Paleozoic magnetizations (e.g., Ku et al., 1967; Spall, 1968) in granites. Spall (1968) reported a paleopole from granite plutons in the Wichita Mountains that was interpreted to be Cambrian in age. Vincenz et al. (1975) reported data from a few sites in granite that yielded Cambrian magnetizations, but most sites yielded magnetizations of late Paleozoic age. Elmore et al. (1998) obtained an interpreted primary paleopole from the Cambrian Colbert Rhyolite in the Arbuckle Mountains, which is part of the Carlton Rhyolite Group (Ham et al., 1964). Price et al. (1998) studied a drill core of the Mount Scott Granite and reported that the magnetic susceptibility increased by about a factor of 50 in the first 30 m, with a corresponding change in color and fracture density. They attributed this to alteration by groundwater, either recent or Permian in age.

METHODS

All samples examined in the present study were obtained from the Martin-Marietta Materials Snyder Quarry (lat. 34.65°N, long. 99.05°W), west of the town of Snyder, Oklahoma, on the west side of Long Mountain (Fig. 1). The quarry is the best exposure of the green Long Mountain Granite (Hamilton et al., 2014). Six sites were collected in green granite, four in red granite, and one in the transition zone between red and green granite. Both the green and red granites were sampled in three areas and depths within the quarry. The presence of a 70-cm-wide diabase dike in the field area permitted a baked contact test. A site (eight samples) was obtained from the diabase itself, and nine cores were collected in a traverse extending from 10 cm to 140 cm (two dike widths) perpendicular to the margin of the dike.

For paleomagnetic study, 2.5-cm-diameter cores were collected from in situ rocks using a portable gasoline-powered water-cooled drill. Specimens were oriented in the field using an inclinometer and Brunton compass. Cores were cut to standard 2.2 cm lengths using a dual-blade water-cooled rock saw. The magnetic susceptibility (MS) and anisotropy of magnetic susceptibility (AMS) were measured using an AGICO MFK-FA1 Multifunction Kappabridge. The AMS data were analyzed using the Anisoft 4.2 software released by AGICO and written by M. Chadima and V. Jelinek. All calculations are done following Jelínek (1978).

The natural remanent magnetization (NRM) was measured using a 2G-Enterprises cryogenic magnetometer with a direct current superconducting quantum interference device housed in a magnetically shielded room. Specimens were then subjected to one to two treatments of low-temperature demagnetization (LTD) in liquid nitrogen to remove the effects of multidomain magnetic minerals (Dunlop and Argyle, 1991; Dunlop et al., 1997; Borradaile et al., 2004). A few pilot specimens were subjected to alternating field (AF) demagnetization up to applied fields of 120 mT (Fig. DR1, a and b[1]) but showed little decay after 30 mT (green granite) or no significant decay (red granite). As a result, most specimens were stepwise thermally demagnetized in an ASC Scientific Thermal Specimen Demagnetizer up to a temperature of

[1]GSA Data Repository item 2015232, tables: major element composition (Table DR1), trace element composition (Table DR2), quantitative compositional data from feldspars (Table DR3) from red and green Long Mountain samples as well as a table (Table DR4) with the green Long Mountain Granite AMS axes; figures; alternating field decay of red and green granite specimens (Figure DR1), low temperature and thermal decay plots for red and green specimens (Figure DR2), magnetic susceptibility versus temperature plots for red and green specimens (Figure DR3), and an equal area plot of the ChRM directions for SDH (dike contact test) specimens (Figure DR4), is available at http://www.geosociety.org/pubs/ft2015.htm or by request to editing@geosociety.org.

700 °C. Demagnetization data were plotted on Zijderveld (1967) diagrams and analyzed using principal component analysis (Kirschvink, 1980); directions with mean angular deviations (MAD) greater than 15° were not used in the analysis. Site statistics are after Fisher (1953).

Hysteresis data were collected from representative samples of both the red and green granite using a Princeton Scientific vibrating sample magnetometer (VSM) at the Paleomagnetism Laboratory at the University of Texas–Dallas. The sample measurements were conducted with a 0.1–0.5 s averaging time and 5 mT steps up to 0.5 T. Corrections were made to the hysteresis loops to remove the paramagnetic component. Magnetic susceptibility versus temperature measurements were acquired using a MK1-FA susceptibility unit interfaced with a CS-4 furnace.

Fifteen polished petrographic thin sections were prepared from the chips cut from the cores, and additional thin sections were provided by Dr. David London. Thin sections were examined using a petrographic microscope in order to determine overall mineralogy and texture of the rock itself as well as to identify magnetic minerals and secondary phases.

Electron microprobe study was performed at the University of Oklahoma using a CAMECA SX50 electron probe microanalyzer. Basic petrographic study was done by backscattered electron imaging (BSEI) coupled with energy-dispersive X-ray analysis (EDXA) using 20 kV accelerating voltage and 10 nA beam current. Quantitative analyses were performed (Pouchou and Pichoir, 1985). Feldspars were analyzed for Na, K, Al, Si, Ca, and Fe using 20 kV by wavelength-dispersive spectrometry (WDS), and matrix corrections used the PAP method acceleration, 10 nA sample current, and counting times of 30 s on peak; a defocused 20 µm spot and continuous transects chosen using BSEI were used in order to integrate bulk compositions. Oxides were analyzed for Si, Ti, Al, Mn, Mg, Cr, Zn, and Fe^{2+} using 20 nA sample current and 2 µm spot size. The identity (oxidation state) of Fe-Ti oxides was derived from electron microprobe analysis, with all Fe analyzed as Fe^{2+} (FeO*) and oxygen determined by stoichiometry. The oxide species was determined by comparing the analytical totals of ($FeO^* + TiO_2$) to the ratios of the masses for stoichiometric oxides with Fe as FeO* divided by their true gram formula weights (George Morgan, 2015, personal commun.). For example, magnetite, Fe_3O_4, contains 3 Fe atoms, yielding a mass of 3 FeO complexes = 215.55 g. Magnetite, however, contains 4 oxygen atoms, yielding a gram formula weight of 231.55 g; hence, this would yield a 93.09 wt% total with Fe analyzed as FeO*. Similarly, hematite would yield an 89.98 wt% total, and goethite would yield a 69 wt% total, versus wüstite and ilmenite, which would yield 100 wt% totals. To better constrain mineralogy, quantitative analysis was performed on other silicate phases (pyroxene, amphibole, micas) using similar methods to those used for the oxides. X-ray mapping of Si, Al, Na, K, and Fe in selected sample areas was performed by WDS using a fixed spot and stage raster; element maps were collected as 256 × 256 arrays using a 200 mS dwell time per pixel.

Six representative rock samples (three red, three green) were collected for detailed whole-rock geochemical analysis. Sample preparation and inductively coupled plasma–mass spectrometry analyses were performed by Activation Laboratories (ActLabs) of Ancaster, Ontario (www.actlabs.com). Fluorine was analyzed by selective ion electrode methods. Ferric iron was measured by titration. Other major and minor elements and oxides were determined by mass spectrometry.

RESULTS AND INTERPRETATIONS

Petrology and Geochemistry

The red granite is exposed in the upper and outer parts of the quarry and is a carapace around the green granite, with the red alteration penetrating farther inward along fractures in places. The red granite is more heavily fractured than the green granite. Epidote-coated fracture surfaces are present in the red granite but not in the green granite. Several diabase dikes (<1 m in width) cut the granite; in the red granite, the dikes are fractured, whereas in the green granite, they tend to be massive. Other studies of the dikes in the granites indicated they are older than the late Cambrian because they do not extend into the Upper Cambrian sedimentary rocks, and some dikes are contemporaneous with the felsic magmatism (e.g., Hogan and Gilbert, 1998).

The red granite is depleted in SiO_2, MnO, and K_2O and enriched in volatiles relative to the green granite, though the differences are small (Table DR1 [see footnote 1]). There is a change in oxidation state of iron (FeO vs. Fe_2O_3), with the average Fe^{2+}/Fe^{3+} dropping from 1.74 to 0.54 between the green and red granites. The total amount of iron, however, is essentially identical (Table DR1 [see footnote 1]). On the whole, the major-element composition of the rock indicates that it is a typical alkaline A-type granite (Bonin, 2007). Trace-element data show that there is no difference between green and red granite, and data are similar to the Mount Scott Granite (Fig. 2; Table DR2 [see footnote 1]).

Green Granite

The Long Mountain Granite is a fine-grained microporphyry, with average size of feldspar and quartz grains ~1–2 mm. The rock is dominantly (>90%–95%) composed of quartz and alkali feldspar intergrowths in a granophyric whole-rock texture (Fig. 3A). Accessory minerals occur as submillimeter crystals in small clusters that may or may not be monomineralic, giving the rock a cumulophyric to glomerophyric texture in addition to the dominant granophyre. Feldspars are also present as exsolved anorthoclase phenocrysts. These sparingly contain partially resorbed plagioclase cores of compositions as calcic as An_{21}. These anorthoclase grains uncommonly have inclusions of other mineral grains and almost always have a potassium-rich rim.

The feldspars in the rock are, with the exception of the rare plagioclase cores, exsolved alkali feldspars. Both perthite and antiperthite crystals are common. Most feldspars are finely

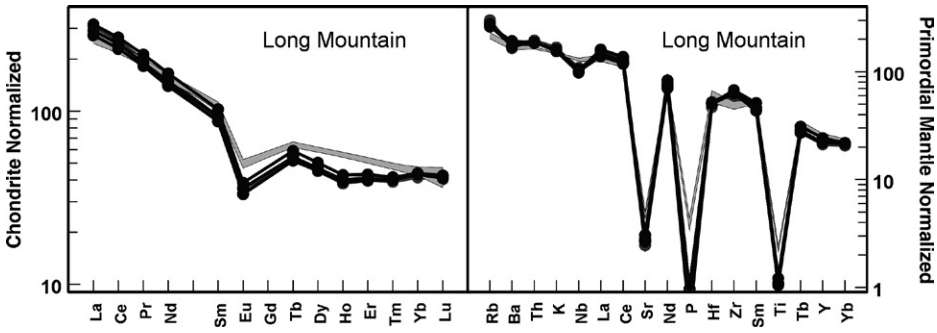


Figure 2. Normalized rare earth element (REE) and trace-element abundances of Long Mountain Granite, with Mount Scott Granite. Long Mountain Granite (red and green) analyses are represented by black dots and lines and are all on top of each other. The gray shading shows Mount Scott Granite data (from Weaver and Gilbert, 1986).

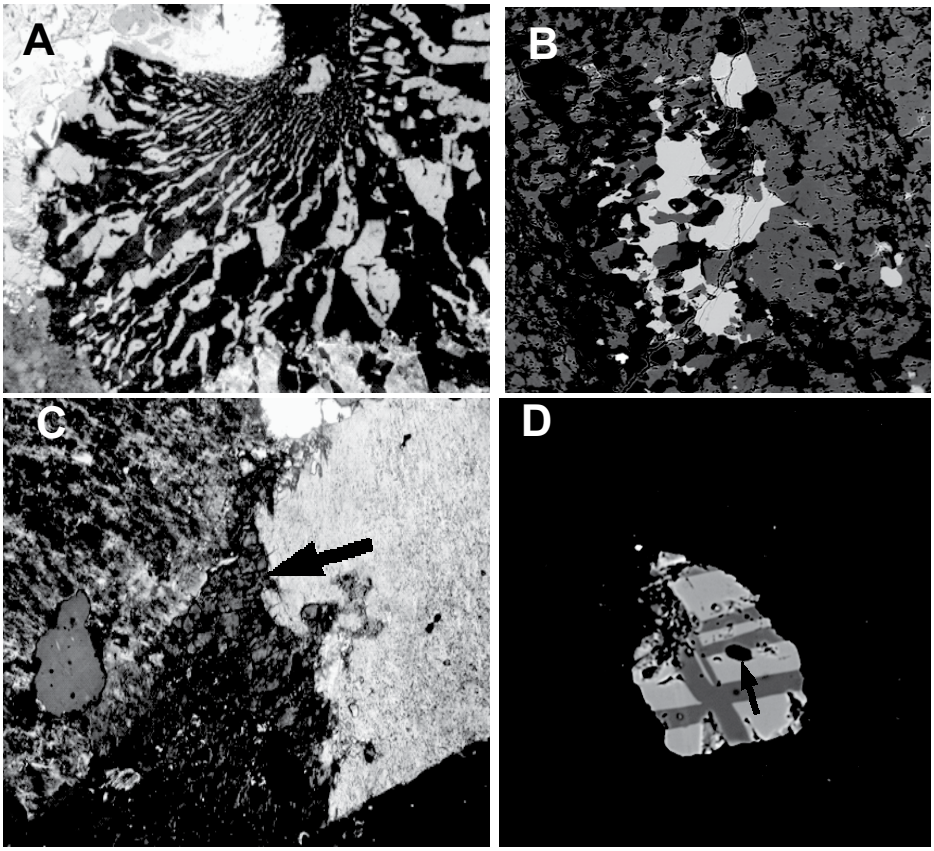


Figure 3. Photomicrographs of green granite. (A) Granophyric quartz-feldspar intergrowth. Crossed polarizers, 3.5 mm field of view. (B) Patchy exsolution in alkali feldspars, backscattered electron image, 790 µm field of view. Bright phase is hornblende. (C) Note mineral in the center (arrow) that appears to be a replacement product of biotite. Crossed polarizers, 1.75 mm field of view. (D) Magnetite-ilmenite grain showing exsolution with an apatite inclusion (indicated by arrow). Backscattered electron image, 158 µm field of view.

exsolved, although some exhibit patchy exsolution (Fig. 3B). Accessory minerals in the granite include a calcic clinopyroxene, end-member hedenbergite ($CaFeSi_2O_6$). Hornblende is sparingly present as anhedral masses in some samples and absent in others. Biotite is also sparingly present, although a brown phase, possibly clay intergrown with Fe hydroxide or Fe oxide, appears to be an alteration product of biotite (Fig. 3C), indicating that biotite may have once been a more abundant mineral. The alteration product is reddish-brown color, has a micaceous habit, is easily damaged by steel needle, and contains small rounded grains of biotite. Epidote-group minerals, identified by microprobe as compositionally similar to allanite, are also present. The most abundant nonsilicate accessory minerals in the Long Mountain Granite are Fe-Ti oxides. Magnetite and ilmenite are present as exsolved grains (Fig. 3D). Fluorite is frequently present as small grains. Other accessory minerals include zircon, monazite, bastnaesite, and chevkinite. Rare grains of apatite have only been found as cores of monazites or as wholly enclosed inclusions in iron oxides.

Red Granite

The red granite is very similar to the green granite in that it consists mainly of granophyric intergrowths. The feldspars also exhibit evidence of substantial alteration such as hematite staining, sericitization, and evidence of recrystallization associated with fractures (Fig. 4A). Authigenic hematite is pervasive in the rock, occurring in hematite-filled fractures, at crystal boundaries, and along exsolution lamellae (Figs. 4A and 4B). Feldspars in red granite have patchy exsolution textures, and some are strongly turbid in addition to being red (Fig. 4C).

Compared to the green granite, many mafic minerals in the red granite are completely altered to a clay phase (Fig. 4D). In thin section, these are typically scoured out in the polishing process, leaving behind void spaces. In some cases, the presence of secondary phases such as epidote-group minerals, calcite, and amorphous silica suggests that mafic minerals may have been dissolved or replaced during diagenesis.

Fe-oxide grains in the red granite tend to appear slightly red in thin section and are interpreted to be oxidized to hematite. Quantitative analysis using WDS on the electron microprobe shows clear clustering of Fe-Ti oxide compositions, one cluster for each color of granite. The cluster for red granite is very close to the ideal for hematite; that for green granite is close to the ideal for magnetite (Fig. 5). Deviations may be due to substituted oxides of trace metals, hydrated minerals, void space, or some combination of these and possibly other factors.

Quantitative analyses of iron concentration (calculated as Fe_2O_3) were performed along transects in anorthoclase and granophyre feldspars. Iron is consistently higher in granophyric feldspar than in anorthoclase, and it is consistently higher in the feldspars of red granite (mean = 0.90%, standard deviation [S.D.] = 0.46, $n = 4$) than in those of green granite (mean = 0.47%, S.D. = 0.26, $n = 4$; Table DR3 [see footnote 1]).

X-ray mapping of element concentrations in 351 × 351 µm areas revealed little detectable iron in the feldspars of green granite, whereas in red granite, count rates can exceed 1000 counts per second (Figs. 6A and 6B). Iron abundance in the feldspars of the red granite appears to correlate roughly with increased sodium (compare Figs. 6B and 6C) and decreased aluminum concentrations (Figs. 6B and 6D), possibly indicating some structural substitution in recrystallized feldspars. BSEI also shows numerous high-scattering threadlike phases in the feldspars, which are consistent with hematite along exsolution planes or cleavage planes.

Anisotropy of Magnetic Susceptibility

The green granites have higher bulk MS (mean = 2.62×10^{-2} SI, S.D. = 5.21×10^{-3}, $n = 43$) than the red granite specimens (MS mean = 3.57×10^{-4} SI, S.D. = 2.74×10^{-4}, $n = 36$). In the green granite, the minimum susceptibility axis (K_3) is vertical or near vertical (Fig. 7; Table DR4 [see footnote 1]) with a slight tilt (11°E) in most sites, whereas the maximum susceptibility axis (K_1) is mostly east-west, and intermediate (K_2) axes are mostly north-south, defining a general subhorizontal or oblate fabric. We note that there is some variation in the K_1 and K_2 axes. Variation has also been reported in some other granites with oblate fabrics (e.g., Ellwood and Whitney, 1980). The green granite has a higher MS, as well as a higher degree of anisotropy than the red granite (Fig. 7B). This oblate fabric in the green granite may represent a primary magmatic

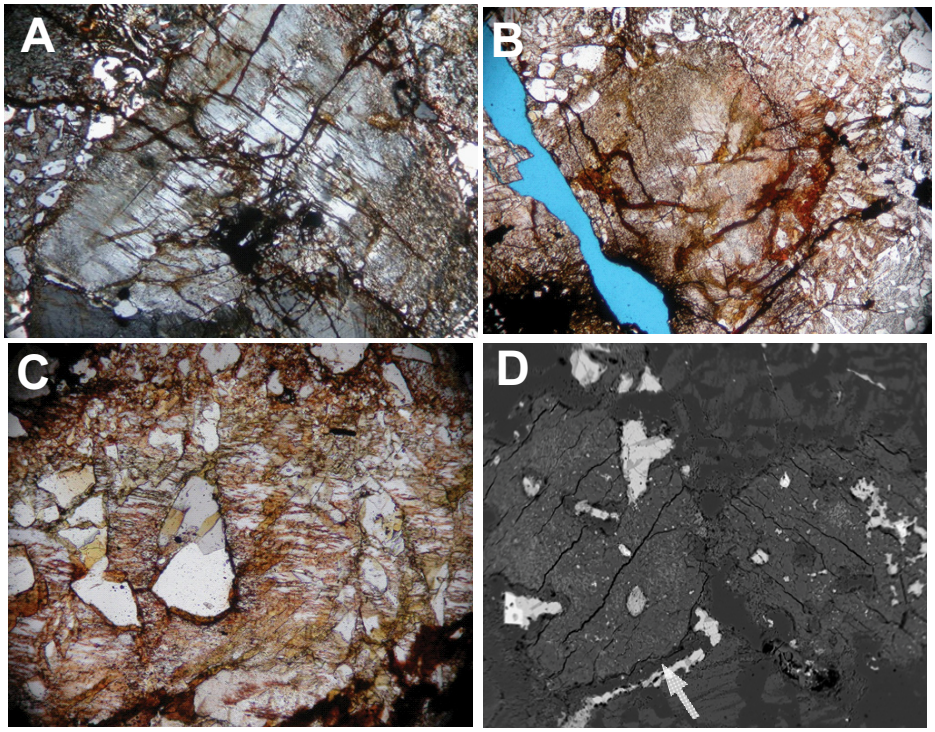


Figure 4. Photomicrographs of red granite. (A) Feldspar crystal showing hematite staining, sericitization, and recrystallization along fractures. 3.5 mm field of view, cross-polarized light. (B) Turbid red-stained feldspar. Plane-polarized light, 3.5 mm field of view. (C) Hematite staining permeating along exsolution lamellae. Plane-polarized light, 1.75 mm field of view. (D) Mafic grains altered to clays (arrow). Backscattered electron image, 1580 µm field of view.

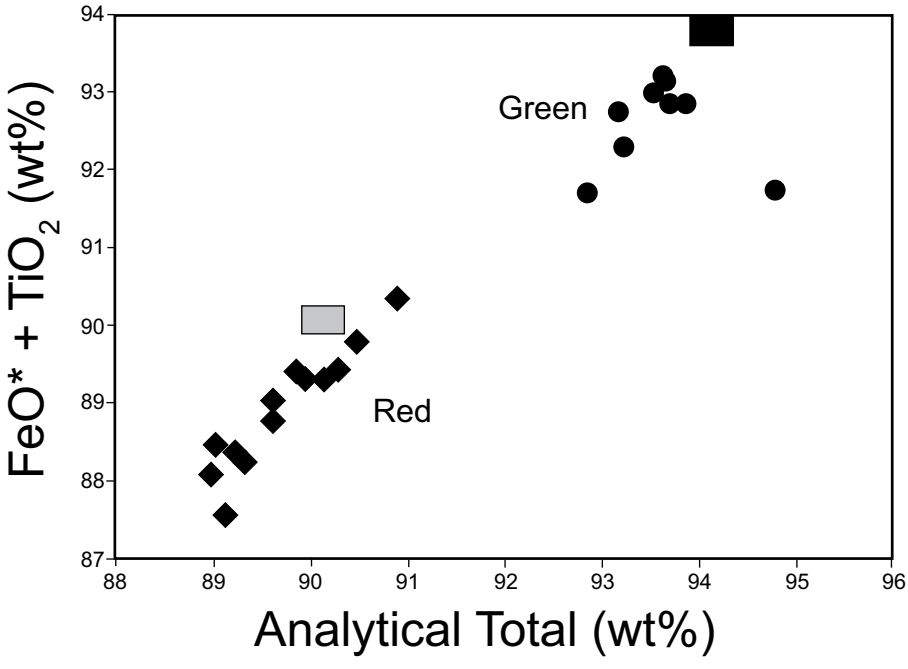


Figure 5. Quantitative wavelength-dispersive spectrometry (WDS) analyses of Fe-Ti oxide compositions. The ideal compositions for hematite and magnetite are indicated by gray and black boxes, respectively. Calculations were done assuming all iron as Fe^{2+}.

fabric, as the Wichita granites are interpreted as large sills, and K_3 is commonly normal to the plane of sill-like bodies when primary fabric is preserved (Ellwood and Whitney, 1980; Rochette et al., 1991, 1992; Ferré et al., 1999).

The MS fabric of the red granite (Fig. 8; Table DR4 [see footnote 1]) is different than that of the green granite. None of the axes is nearly as vertical as K_3 in the green granite (Fig. 8). Magnetic lineation (K_1) is north in all sites, with K_2 and K_3 alternating between northeast-southwest and southeast directions. The red granite does not reveal any particular shape parameter, in contrast to the oblate fabric in the green granite (Fig. 7C). The fabric in the red granite is not interpreted to represent a magmatic fabric.

Paleomagnetism

The green granites have higher natural remanent magnetization (NRM) intensity (mean = 3.15×10^2 mA/m, S.D. = of 3.69×10^2, $n = 43$) than the red granite specimens (mean = 3.59 mA/m, S.D. = of 1.67, $n = 36$).

Green Granite

LTD treatment prior to thermal demagnetization of green granite specimens resulted in a considerable decrease in magnetization relative to the NRM, averaging ~71% (S.D. = 6%; Fig. DR2a [see footnote 1]). This suggests the presence of multidomain magnetite. The direction removed has steep positive inclinations and generally north declination that is similar to the modern direction (declination = 5.5°E, inclination = 63.5°).

During thermal demagnetization, all specimens showed complete unblocking below 580 °C (Figs. 9A and 9B; Fig. DR2a [see footnote 1]), indicating that the remanence is likely dominated by magnetite. Specimens demagnetized by AF treatment show considerable decay by 30 mT and complete randomization by 100 mT (Fig. DR1a [see footnote 1]), which is consistent with magnetite as the carrier of the remanence. Hysteresis measurements (Fig. 10A) for representative samples are consistent with the interpretation that the magnetization resides in magnetite in the green granite. The samples reach saturation by ~300 mT (Fig. 10A). Compared to the red granite samples (Fig. 10B), the saturation magnetization (Ms) and saturation of remanence (Mr) are much higher in the green granite. The coercive force (Hc) and coercivity of remanence (Hcr) are 3.2–5.0 mT and 15.2–29.3 mT, respectively, which are within the range for magnetite (Hc: 5–80 mT; Hcr: 15–100 mT; Dekkers, 2007). Magnetic susceptibility versus temperature measurements

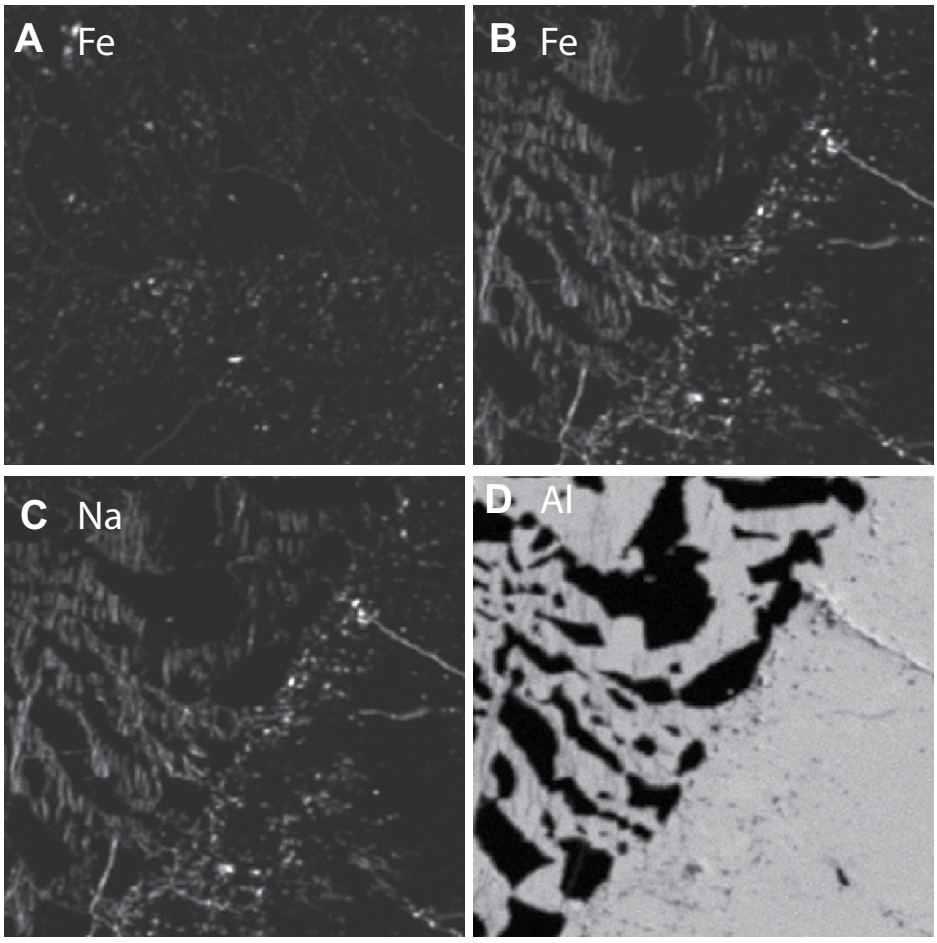


Figure 6. Elemental abundance maps for Fe, Na, and Al. (A) Green granite (SG 2–1). (B–D) Red granite (SR 1–3). All images have a 351 µm field of view.

shows a steep decline at 580 °C (Fig. DR3a [see footnote 1]), also indicating magnetite.

Thermal demagnetization of green granite specimens removes a characteristic remanent magnetization (ChRM) of ESE declination (D) and moderate positive inclination (I). The mean for five sites is D = 104.6°, I = 54.1°, α_{95} (cone of 95% confidence) = 11.5°, *k* (precision parameter) = 45.4 (Figs. 9A and 9B; Table 1). A modern viscous remanent magnetization (VRM) is not typically isolated in thermal demagnetization, which is consistent with its removal by LTD. The ChRM is unblocked in most specimens by 500 °C but can persist up to 520–540 °C in some specimens. Mean MAD values for this component are typically less than 10°.

Directions of magnetization in the green granite sites are moderately to well grouped (α_{95} = 11.5; Fig. 11; Table 1). Site SG 3 is divided into two groups, one with a mean direction similar to those of the other sites, and one with a more northerly declination and steep positive inclination similar to the modern direction. Site SG 4 also exhibits a north-directed and steep positive inclination site mean. Therefore, SG 4 and the second group from SG 3 were not used in calculating the overall mean direction. The pole for the green granite is 9.0°N, 314.6°E (A_{95} = 15.0; Fig. 12; Table 1).

A contact test involved sampling an ~70-cm-wide dike (SDD) and sampling in the green granite progressively outward up to 1.4 m from the edge of the dike (SDH). Specimens from the dike did not yield a stable magnetization, and thermal decay profiles show variable laboratory temperatures and sporadic behavior. This is interpreted as being due to the formation of new magnetic minerals during thermal demagnetization. Microprobe inspection indicates the presence of biotite, pyrite, and ferrosphalerite, which break down to iron oxides during heating, in addition to possible oxidation of primary titanomagnetite and ilmenite. No consistent difference or trend in magnetization directions was observed between specimens near the dike versus those farther away (Fig. DR4 [see footnote 1]). This indicates a failed contact test. The mean direction from site SDH (Table 1) was included in the calculation of the overall mean for the green granite.

Red Granite

Specimens of red granite from sites SR 1, ST 1, and ST 2 were subjected to one LTD treatment, and specimens from sites SQR 1 and SQR 2 were subjected to two LTD steps (Fig. DR2b [see footnote 1]). The second treatment did not significantly change the magnetic intensity (Fig. DR2b [see footnote 1]). Some specimens exhibited very little NRM change, as low as 2% after LTD, while others lost up to 62% of the NRM intensity. Some specimens from sites SQR 1 and SQR 2 increased in intensity during LTD treatment. This is interpreted to have been caused by the removal of the north-directed VRM, which uncovered the south-directed ChRM. The average change in intensity was 27% (S.D. = 18%). The results from specimens that showed significant decay suggest the presence of some multidomain magnetite in the specimens, although not as much as in the green granite.

A modern VRM, possibly in goethite, was removed by 100–200 °C. Further thermal demagnetization of specimens from red granite sites removed a ChRM with southeast declination and shallow positive inclinations from 300 °C to 680 °C (Table 1; Figs. 9C and 9D).

Decay above 580 °C and up to 680 °C indicates a significant contribution from hematite. The AF treatment of red specimens produced limited decay (Fig. DR1b [see footnote 1]), and the lack of complete or significant decay is consistent with the presence of abundant hematite. The hysteresis measurements are consistent with the presence of hematite (Fig. 10B). The Ms and Mr values are lower and the Hc and Hcr are higher than for the green granite (Fig. 10). A hysteresis loop from a red sample was saturated by ~500 mT (Fig. 10B), indicating the presence of both magnetite and a higher-coercivity magnetic phase. Magnetic susceptibility versus temperature measurements show a steep decline at 580 °C (Fig. DR3b [see footnote 1]) similar to the green granite. This suggests that the magnetite grains are producing most of the response to magnetic susceptibility measurements.

Magnetizations isolated at sites in red granite are well grouped (Fig. 11; Table 1), except for site ST 1, which was collected in the transition zone between red and green granites and displays characteristics similar to the green granite. Consequently, ST 1 was not used in computation of the mean direction of the red granite (D = 150.6°, I = 2.1°, α_{95} = 8.3°, *k* = 123.8). The pole for the red granite is 44.9°S, 304.9°E (A_{95} = 6.2; Fig. 12; Table 1).

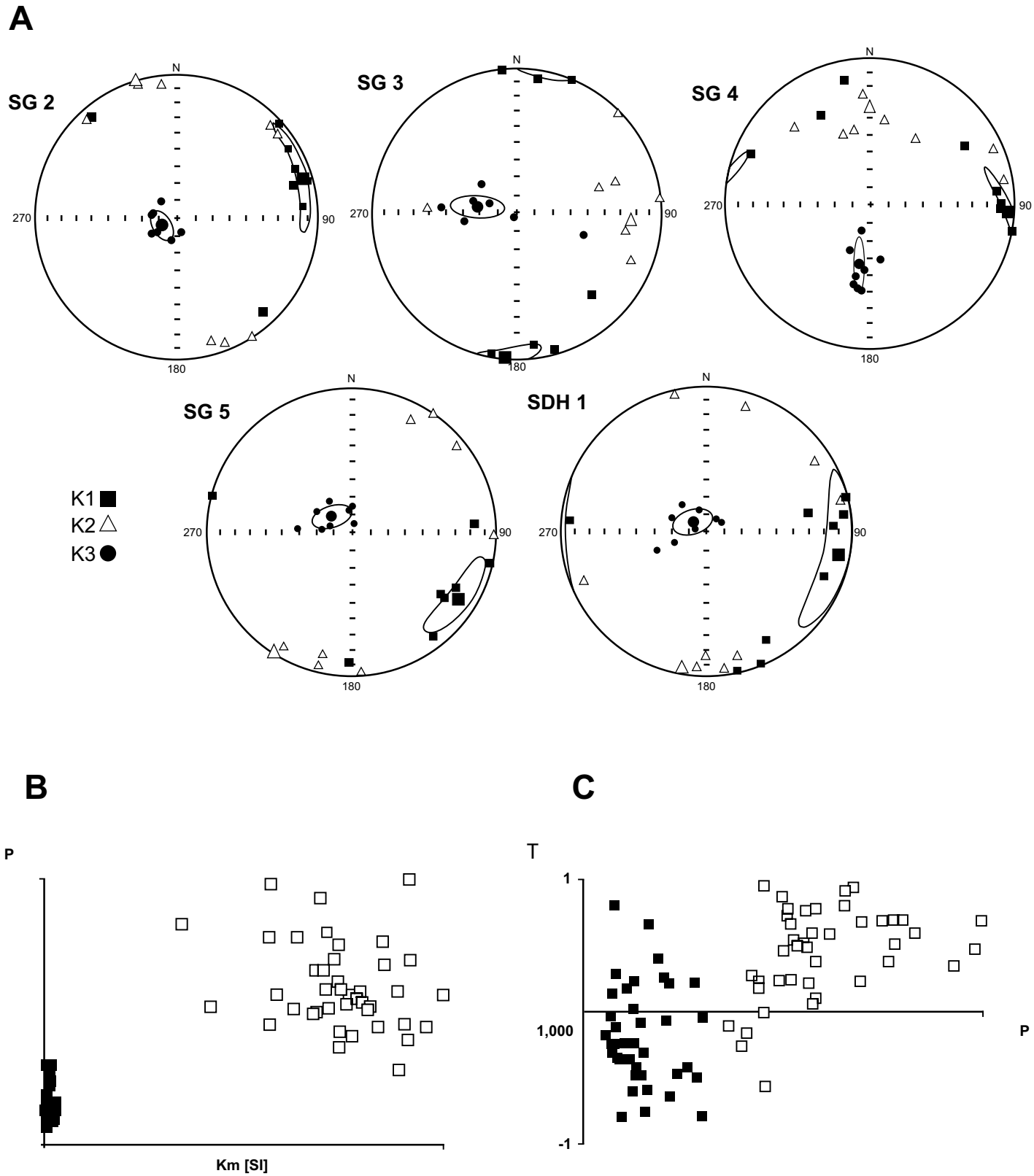


Figure 7. (A) Principal axes of magnetic susceptibility in green granite sites. Larger symbols are means in A. (B) Magnetic anisotropy (P) versus maximum susceptibility. (C) Shape parameter ($T = 2\ln[K_2/K_3]/\ln[K_1/K_3]$; Tarling and Hrouda, 1993) versus P. In B and C, closed symbols indicate red granite, and open symbols are green granite.

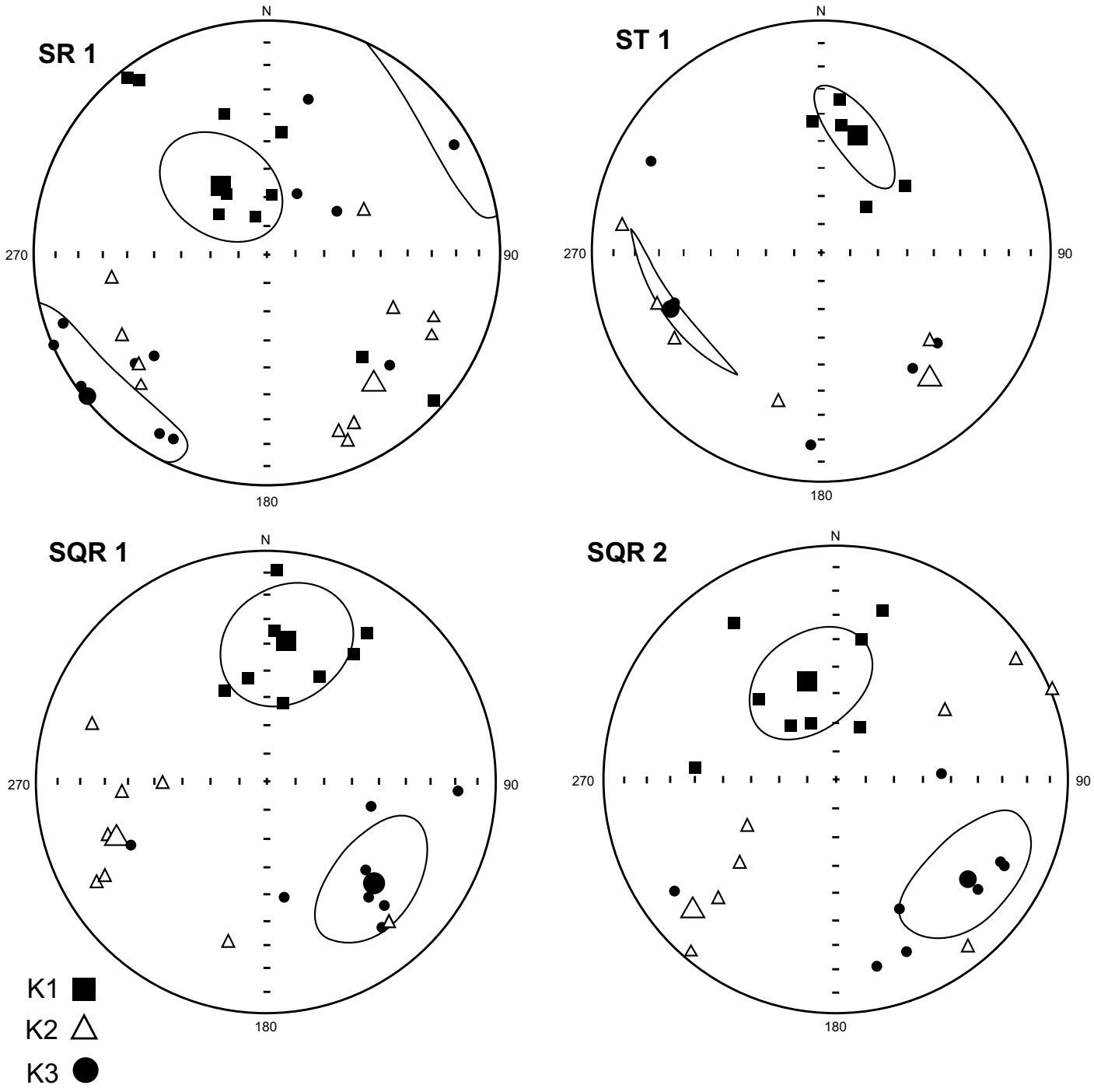


Figure 8. Principal axes of magnetic susceptibility in selected sites of the red granite.

DISCUSSION

Green Granite

The Long Mountain Granite was emplaced in Cambrian time, presumably around the time of the 533 Ma Mount Scott Granite (Wright et al., 1996). It may crosscut the Mount Scott Granite (Myers et al., 1981) and therefore be younger. As previously stated, the Wichita Granites were emplaced as shallow sill-like bodies (e.g., Hogan et al., 2000). The subhorizontal magnetic fabric of green facies rocks of the Long Mountain Granite is consistent with this emplacement mode.

There are limited numbers of sites, and all the sites were collected from one quarry, which raises the question of whether secular variation was averaged. The sites were collected from three different locations and depths within the quarry. In addition, Price (2011) estimated that the Mount Scott Granite cooled to 500 °C in 1 m.y. Even if the Long Mountain Granite cooled at a faster rate, there was probably more than enough time to average secular variation.

The pole position defined by the ChRM of the green granite (9.0°N, 314.6°E) plots near the Ordovician part of the APWP by Torsvik et al.

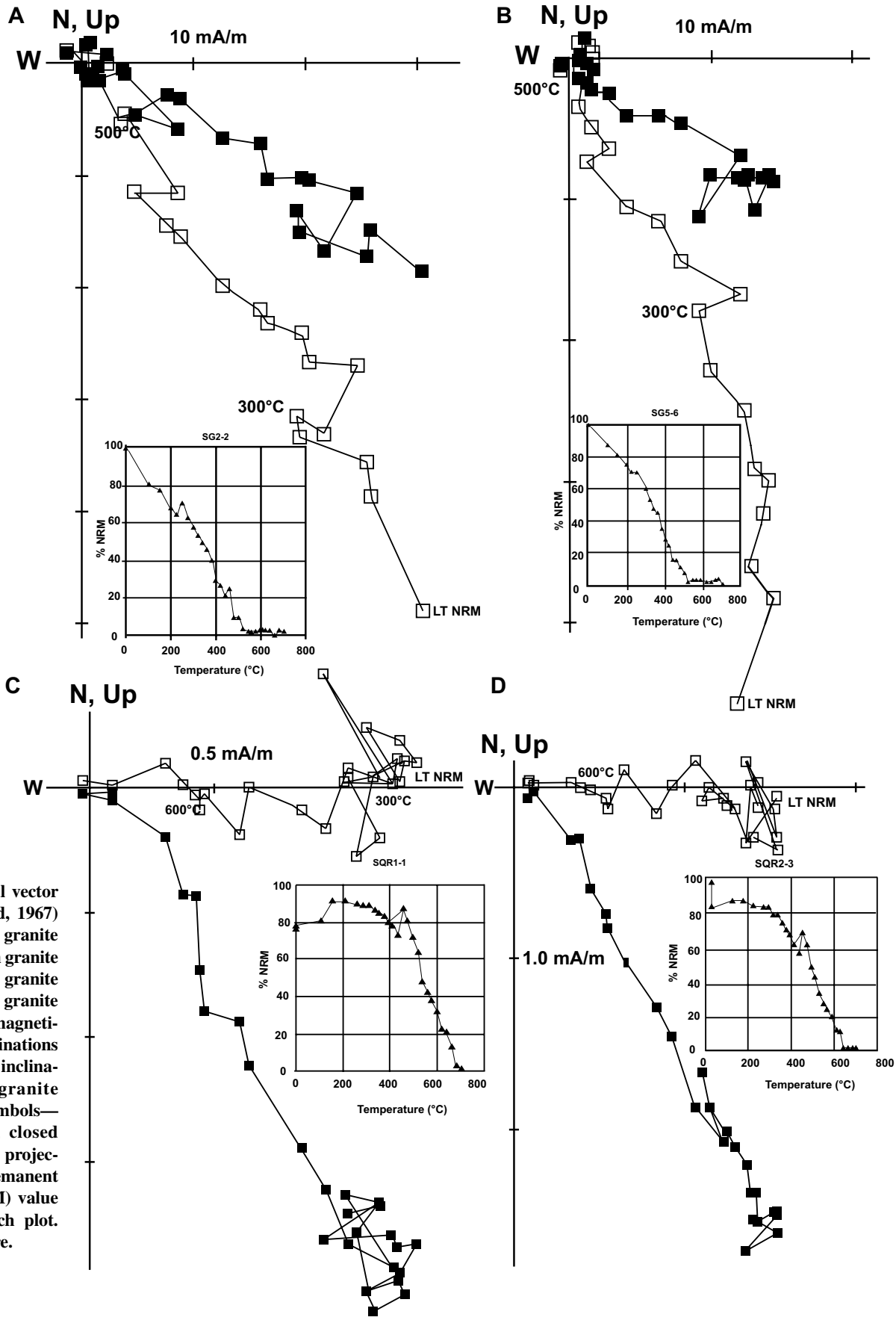


Figure 9. Orthogonal vector diagrams (Zijderveld, 1967) for green and red granite specimens. (A) Green granite (SG 2–2). (B) Green granite (SG 5–6). The green granite specimens yield a magnetization with ESE declinations and moderate down inclinations. (C–D) Red granite specimens. Open symbols—vertical projection; closed symbols—horizontal projection. The natural remanent magnetization (NRM) value is not shown in each plot. LT—low temperature.

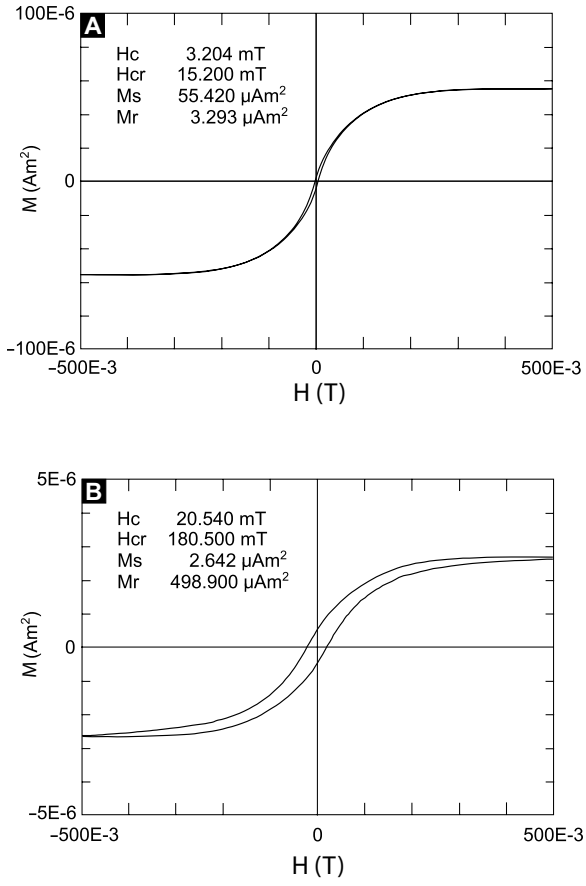


Figure 10. Hysteresis loops from samples of the (A) green granite and (B) red granite.

(2012) (Fig. 12). The pole is consistent with other paleomagnetic poles from rocks of early Paleozoic age in southern Oklahoma, including those of Spall (1968), who reported a paleopole from the Wichita Granite Group at 2°N, 147°E, which was interpreted to reside in titanomagnetite. Elmore et al. (1998) reported an apparent preliminary pole (11°S, 151°E) from the Colbert Rhyolite (an extension of the Carlton Rhyolite Group) in the Arbuckle Mountains.

There is no evidence that the Long Mountain Granite was significantly tilted after emplacement, which suggests that paleohorizontal is preserved. We note, however, that some of the granites in the Wichita Mountains, particularly in the east, are gently folded (~10°). The axis of the closest fold is 9 miles (14.5 km) north of Long Mountain (Gilbert, 1982). The deformation is interpreted to Pennsylvanian in age (Gilbert, 1982).

The pole position for the green granite lies to the west of the 530 Ma segment and near the Ordovician part (Fig. 12) of the APWP proposed by Torsvik et al. (2012). There are several possible explanations for this. The Long Mountain Granite has been interpreted as Cambrian in age, although it is assumed to crosscut the Mount Scott Granite (Myers et al., 1981) and is therefore younger. It is not likely to be Ordovician; there is no other evidence for Ordovician igneous activity in the Southern Oklahoma aulacogen. Another possibility is that the ChRM in the green granite could have been acquired during hydrothermal alteration, although there is no other evidence for hydrothermal events of Ordovician age in the Southern Oklahoma aulacogen.

The thermal unblocking of the remanence in green granite specimens by 520 °C, randomization in AF demagnetization to 100 mT (Dunlop et al., 1997; Borradaile et al., 2004), the rock magnetic results, and the presence of magnetite all provide evidence that the ChRM in the green granite resides in magnetite. The oblate magnetic fabric in the green Long Mountain Granite is interpreted to be primary, and it is consistent with a primary magnetization. Other studies have interpreted similar fabrics as primary in intrusive rocks (e.g., Ferré et al., 1999; Stevenson et al., 2007).

TABLE 1. PALEOMAGNETIC DATA FOR SITES IN THE GREEN GRANITE AND RED GRANITE, LONG MOUNTAIN GRANITE, OKLAHOMA

Green Granite									
Site	N/N_0	Dec	Inc	α_{95}	k	Latitude (°N)	Longitude (°E)	K	A_{95}
SG 1	5/7	94.1	59.9	8.6	81	19.1	314.1		
SG 2	7/8	108.3	47.8	9.2	43.6	3.3	316.7		
SG 3a	4/7	129.7	50	19.3	23.6	–9.2	303.1		
SG 3b*	3/7	12.5	74.3	11	127.3	62.8	274.5		
SG 4*	6/8	21.2	64.2	13.2	26.8	70.4	309.5		
SG 5	7/8	82.5	48.7	14.1	19.3	22	329.4		
SDH	8/9	107.2	57.4	19.1	9.4	9.1	310.7		
Mean	—	104.6	54.1	11.5	45.4	9.0N	314.6E	26.8	15.0
Red Granite									
Site	N/N_0	Dec	Inc	α_{95}	k	Latitude (°N)	Longitude (°E)	K	A_{95}
SR 1	11/11	156.5	10	8.6	26.4	–44.5	294.9		
ST 1†	3/7	144.4	58.3	11.3	120.1	–9.3	288.3		
ST 2	3/3	152.1	1.8	15.5	6.6	–45.9	303.3		
SQR 1	8/8	146.3	–0.7	7.9	50.7	–43.5	310.9		
SQR 2	7/8	147.6	–2.9	6.9	78.11	–45.1	310.4		
Mean	—	150.6	2.1	8.3	123.8	44.9S	304.9E	223	6.2

Note: N/N_0—number of specimens accepted for statitical purposes vs. number of specimens demagnetized; Dec—declination; Inc—inclination; a_{95}—95% cone of confidence; k—Fisher's (1953) best estimate of the precision parameter; K—best estimate of the precision paramater for the mean pole; A_{95}—95% confidence circle around the pole. Units for Dec, Inc, and α_{95} are degrees.

*SG 3b and SG 4 not used in calculation of overall average.

†Site ST 1 was not included in calculation of mean. Site SDD, from a dike, did not yield a stable magnetization, and thermal decay profiles show variable laboratory temperatures and sporadic behavior.

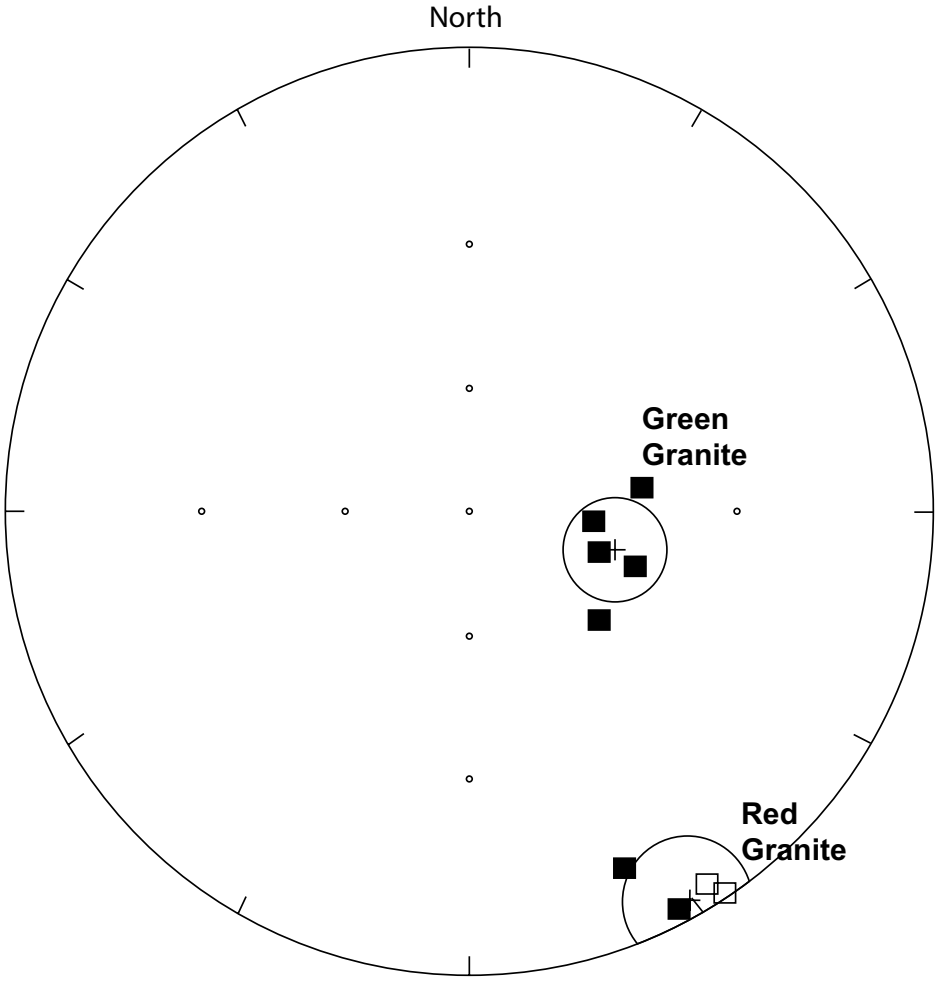


Figure 11. Equal-area projection of characteristic remanent magnetization (ChRM) site mean directions from green and red granite. There is no overlap between the two populations. Open symbols—negative (up) inclination; solid symbols—positive (down) inclinations. The circles are the projected cones of 95% confidence for the means of each grouping.

One argument against a primary remanence in the green granite is the failed contact test. Heating due to mafic dike intrusion should have been sufficient to reset the magnetization near the dike margin, but this is not the case. In addition, there is no petrographic evidence for alteration that could have reset the magnetization after the emplacement of the dike. Given that most models of the evolution of the Southern Oklahoma aulacogen suggest a short duration of felsic magmatism overlapping with, and followed by, mafic magmatism, it is possible that there may not have been sufficient time between the granite emplacement and the dike intrusion to result in a major change in field direction.

There is evidence that the Long Mountain Granite underwent significant hydrothermal/deuteric modification following emplacement and crystallization. For example, the feldspars in the green Long Mountain Granite contain coarsened and patchy exsolution features (Fig. 3C) of the type that have been interpreted to result from recrystallization of perthite and antiperthite under deuteric and hydrothermal conditions (Parsons, 1978; Worden et al., 1990; Parsons and Lee, 2009). The altered biotites also suggest that the green granite was altered.

The ChRM in the green granite is provisionally interpreted as a primary Cambrian thermal remanent magnetization or an early acquired magnetization related to deuteric/hydrothermal alteration. The APWP for North America during the Ediacaran and Cambrian remains very poorly defined and controversial (e.g., Mitchell et al., 2011). The pole from the Mount Rigaud–Chatham-Grenville stocks (11.9°S, 184.6°E; McCausland et al., 2007) is the only other pole for ca. 530 Ma from Laurentia. The green Long Mountain Granite pole is to the west of the Mount Rigaud–Chatham-Grenville pole. The reason for the difference in the two poles could be that each represents different readings of the ca. 530 Ma paleofield for Laurentia. Further work on the Cambrian rocks in southern Oklahoma is needed to better understand the significance of the Long Mountain Granite pole in terms of Ediacaran and Cambrian apparent polar wander.

Red Granite

The red granite has been extensively altered compared to the green granite. Evidence for alteration includes abundant authigenic hematite, sericitization, and destruction of mafic silicate grains. The red granite contains abundant hematite-filled fractures, and strong hematite staining at crystal boundaries and along exsolution laminae (Figs. 4A and 4B). Extensive feldspar recrystallization in the red granite produced patchy, porous, turbid, red feldspars. Quantitative analysis and elemental abundance mapping indicate that feldspars in the red granite contain significantly more iron than do feldspars in the green granite. The bulk iron abundance is the same in the red and green granite, although the valences are different, suggesting that the iron for the hematite was sourced from the granite itself and was not externally derived.

These occurrences of hematite in the red granite are consistent with data from other workers who have reported hematite in pores in feldspars, which was interpreted to have precipitated during feldspar recrystallization during cooling from magmatic temperatures (Boone, 1969; Putnis et al., 2007). In contrast, other studies have reported recrystallization of K-feldspar and albitization of plagioclase associated with red granites in weathering profiles (Parcerisa et al., 2010; Ricordel et al., 2007). Hematite has also been associated with fractures by other workers (e.g., Drake et al., 2008; Plümper and Putnis, 2009; Sandström et al., 2010).

Void spaces in the red Long Mountain Granite left by destruction of mafic grains are variably filled with calcite and epidote-group minerals. Epidote also coats fractures in the red granite, though it has not been observed in the green granite. Epidote can form under many conditions. For example, Franklyn et al. (1991) ascribed epidote as forming from the dissolution of plagioclase below 70 °C, and Pearson (1987) reported epidote forming from alkali feldspar between 70 °C and 160°C. Ricordel et al. (2007) described epidote in weathered red granites and in association with altered biotite and albitization. Other papers have reported epidote forming at 160 °C (Kerrich and Kamineni, 1988) or at 130 °C (Strens, 1964) or at temperatures of 320 °C or above (Deer et al., 1992).

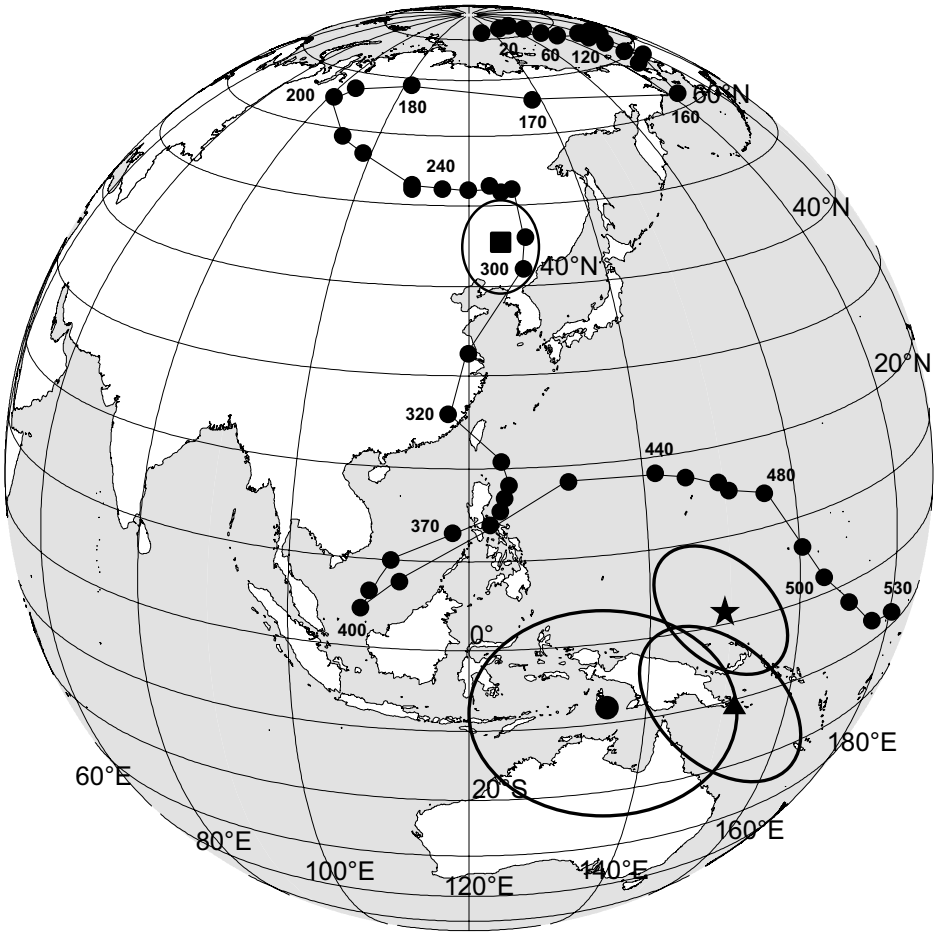


Figure 12. North American Phanerozoic apparent polar wander path (Torsvik et al., 2012) with paleopoles (antipodes from Table 1) and ovals of 95% confidence plotted for data from the green (circle) and red (square) Long Mountain Granite. Triangle is paleomagnetic pole from the ca. 530 Ma Colbert Rhyolite (Elmore et al., 1998), and the star is the paleomagnetic pole from other Wichita Granite Group samples (Spall, 1968).

The red granite has lower bulk MS and NRM than the green granite. These results are consistent with the findings of Price et al. (1998) in their study of effects of weathering on the magnetic properties of the Mount Scott Granite in drill core.

The ChRM of the red granite has a consistent southeast declination and very shallow inclination (Fig. 11). The ChRM is unblocked over a temperature range to 680 °C. This, along with the results of the AF demagnetization, hysteresis measurements, and the petrographic evidence, indicates the ChRM resides in hematite. In order for the ChRM to be a thermoviscous remanent magnetization (TVRM) the rock would need to have been heated above 650 °C according to the time-temperature relations given by Pullaiah et al. (1975). There is no evidence that the Long Mountain Granite was exposed to such high temperatures since emplacement. Given that the dominant sediments and rocks in the area are Permian conglomerates sourced from the Wichita Granite Group, and virtually no tectonism has affected the region since the Permian, the maximum burial depth never exceeded a few kilometers, and burial temperatures never reached 100 °C. We interpret the ChRM as a chemical remanent magnetization (CRM) residing in hematite. The intensity decreases in LTD suggest that magnetite also contributes to the total remanence, but it is not apparent if magnetite also carries a geological ancient remanence in the red granite. The maximum unblocking temperature persists well above the magnetite Curie temperature. The red granite has a low level of MS anisotropy, and the AMS fabric does not have a consistent orientation. Similar to the red Long Mountain Granite, other workers have also reported that red granites have reduced magnetic susceptibility (Eliasson, 1993; Price et al., 1998; Mattsson and Thunehed, 2004). Previous studies of AMS in reddened and oxidized granites have suggested that the primary fabric is not preserved (e.g., Archanjo et al., 2009). The red granite is interpreted to contain an alteration AMS fabric.

The paleopole calculated for the red Long Mountain Granite (44.9°S, 304.9°E) plots near the 290–300 Ma segment of the North American APWP of Torsvik et al. (2012) (Fig. 12). Recent work on the North American APWP for the Paleozoic suggests that many of the poles used to define the path may be inaccurate due to effects of inclination shallowing (Bilardello and Kodama, 2010a, 2010b). The Torsvik et al. (2012) path accounts for this inclination shallowing. One issue that needs to be addressed is the possibility that the hematite CRM directions could have been affected by preexisting mineral fabric (Kodama, 2012). For example, several studies have shown that a CRM in hematite that grew from alteration of preexisting magnetite can have a direction with an intermediate direction between the preexisting magnetite and the magnetic field during growth, and thus it does not reliably record the growth field (e.g., Walderhaug, 1992; Madsen et al., 2002). Stokking and Tauxe (1990a, 1990b) have shown that hematite (single or two generations) grown in one field reliably records the growth field. In contrast, hematite synthesized in multiple generations and acquired in perpendicular fields contains a complex magnetization with intermediate or parallel/antiparallel directions (Stokking and Tauxe, 1990a). We note that much of the hematite in the Long Mountain Granite occurs in fractures and in feldspars not directly associated with primary magnetite and that the red granite CRM displays univectorial decay without parallel or antiparallel directions. Although we cannot rule out the possibility that the hematite was affected by the magnetite magnetization, we suggest that the hematite formed at approximately the same time, and the CRM probably represents the average field during the growth of the hematite.

Hematite authigenesis that resulted in reddening of the Long Mountain Granite is interpreted to be related to low-temperature weathering fluids. The Permian age of the hematite-carried CRM is strong evidence for this interpretation and correlates well with exposure during the Permian (Gilbert, 1982). Mid-Permian alteration is more consistent with low-temperature weathering fluids than with high-temperature fluids such as those that might occur with uplift in the late Mississippian–Pennsylvanian. Also, if late Paleozoic hydrothermal fluids did result in alteration, this should also have affected the green granite, but there is little evidence for such

alteration. Hot fluids would likely have remagnetized the magnetite in the green granite, but there is no evidence for this. The turbid feldspars and the presence of epidote in the red granite could suggest alteration by hydrothermal fluids, although similar alteration can be explained by low-temperature weathering. This interpretation of alteration by weathering fluids is consistent with a study of red Precambrian (ca. 1500 Ma) granite in the St. Francois Mountains of Missouri, which reported enriched O^{18} relative to unoxidized granites (Wenner and Taylor, 1976; Taylor, 1977) and concluded the alteration was due to interaction with low-temperature waters 200–300 m.y. after the emplacement of the granites (Wenner and Taylor, 1976).

The paleomagnetic signature carried by hematite clearly shows that the red coloration in Long Mountain Granite exposures postdates intrusion by ~250–300 m.y. Permian to Triassic CRMs are reported in alteration zones below paleoweathering surfaces in crystalline basement rocks in Europe (e.g., Ricordel et al., 2007; Franke et al., 2010; Fàbrega et al., 2012). The CRMs in hematite are interpreted to have been acquired during weathering and can be related to an increase in chemical weathering at or near the Permian-Triassic boundary (Sheldon, 2006; Martin and Macdougall, 1995). The mechanism of remagnetization in the Long Mountain Granite is similar to that from these Permian-Triassic paleoweathering surfaces in Europe, but the timing is Permian, and it is not related to climate change at the Permian-Triassic boundary. The change from icehouse to greenhouse conditions, however, occurred over 40 m.y. in the Permian and was marked by increases and decreases in carbon dioxide and associated changes in temperature (Montañez et al., 2007), and it is possible that the alteration in the Long Mountain Granite could be associated with one of the periods of increased carbon dioxide in the Permian. The pole plots between 300 and 290 Ma on the APWP (Fig. 12), which does not coincide with an episode of high carbon dioxide, although an episode started at ca. 290 Ma (Montañez et al., 2007). The alteration in the Long Mountain Granite is interpreted to have been caused by local tectonic/weathering events.

CONCLUSIONS

The green facies of the Long Mountain Granite contains a ChRM that resides in magnetite, and the paleopole position is consistent with some other poles for Cambrian igneous rocks of the Southern Oklahoma aulacogen. However, the pole is to the southwest of the Cambrian part of the APWP. The magnetization in the green granite is interpreted as a primary Cambrian thermal remanent magnetization (TRM) or an early magnetization acquired during deuteric/hydrothermal alteration. The oblate AMS fabric is interpreted as primary in origin and is consistent with the interpretation of the magnetization as a TRM. Further work is needed on Cambrian rocks in southern Oklahoma and elsewhere to better evaluate controversial hypotheses (e.g., Kirschvink et al., 1997; Abrajevitch and Van der Voo, 2010; Bono and Tarduno, 2015) regarding Ediacaran and Cambrian apparent polar wander.

The red facies of the Long Mountain Granite is more heavily fractured than the green facies and occurs as a carapace around the sill. The red granite is in bulk very similar to the green granite in terms of minor and trace elements. The red granite does not show significant differences in major-element content except for the preferred oxidation state of iron, with average Fe^{2+}/Fe^{3+} dropping from 1.74 to 0.54. The total amount of iron is essentially identical (iron close to 2 wt%) in both the red and green granite. Ferromagnesian minerals in the red granite are completely altered, and the voids are filled with clays, amorphous silica, calcite, and epidote-group minerals. Most large oxide grains in the red granite have been oxidized to hematite. Feldspar recrystallization in the red granite led to patchy, turbid, red feldspars. The reddening in the feldspars is caused by the hematite, which is dispersed along fractures and exsolution planes. Hematite in pore spaces in altered feldspars may also have contributed to the reddening. Quantitative analysis and elemental abundance mapping indicate that feldspars in the red granite contain significantly more iron than do feldspars in the green granite.

Any magnetic fabric in the green Long Mountain Granite was destroyed in the process of reddening, as shown by the AMS data. Degree of anisotropy decreased to an average of less than 1%, and the fabric is interpreted to have been caused by alteration. The CRM in the red granite resides in hematite and yields a paleomagnetic pole position that lies on the 290–300 Ma part of the APWP.

The results of this work are not consistent with the hypothesis that the red facies of the Long Mountain Granite resulted from deuteric magmatic–hydrothermal system interactions. There is evidence of deuteric alteration in the green granite but no evidence that it resulted in the red coloration of the red granite. In the Long Mountain Granite, and likely in other Wichita granites, hematite responsible for the red coloration postdates the crystallization of the granite by 250–300 m.y. and is interpreted to have formed during low-temperature weathering. The results from the Long Mountain Granite are not consistent with alteration caused by widespread paleoclimatic conditions in the Late Permian and are interpreted as related to local tectonic and/or weathering events.

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